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ORIGINAL ARTICLE

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Garlic (*Allium sativum*) improves anxiety- and depressive-related behaviors and brain oxidative stress in diabetic rats

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ABSTRACT

This study investigated the effects of garlic on anxiety- and depression-related behaviors and brain oxidative markers in streptozotocin (STZ)-induced diabetes in rats. Fifty-six male Wistar rats were randomly divided into seven experimental groups ($n = 8$/group): control, diabetic + saline, diabetic + garlic, diabetic + imipramine, and diabetic + diazepam groups. Animals received garlic homogenate (0.1, 0.25, and 0.5 g/kg) for 10 days. At the end of the treatments, anxiety- and depressive-related behaviors were evaluated by elevated plus maze (EPM) and forced swimming test (FST), respectively. Superoxide dismutase (SOD) and glutathione peroxidase (GPx) activities and malondialdehyde (MDA) levels were measured in the brain. Diabetic + garlic (0.5 g/kg) group showed lower anxiety- and- depressive-like behaviors as compared to the diabetic rats. Furthermore, garlic treatment (0.5 g/kg) attenuated MDA levels and enhanced SOD and GPx activities in the brain. Our findings indicate that garlic alleviates anxiety- and depression-related behaviors in the diabetic rats possibly by attenuation of brain oxidative stress.

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Diabetes; garlic; anxiety; depression; oxidative stress

Introduction

Diabetes is associated with an increased risk of psychological problems such as anxiety and depression (Peyrot and Rubin 1997, Kuhad and Chopra 2008). The prevalence of anxiety and depression symptoms in the diabetic patients is considerably higher than healthy people (Grigsby *et al.* 2002, Collins *et al.* 2009). Different psychiatric disorders particularly anxiety, depression, anger, and sleeping problems, are also associated with an increased risk for the development of type 2 diabetes (Pouwer *et al.* 2010). The presence of anxiety disorder is especially associated with both poorer glucose control and increased clinical symptoms of diabetes (Li *et al.* 2008, Siddiqui 2014). Depression also increases the risk of diabetes complications and results in poor metabolic control and hyperglycemia accentuates depression (Lustman and Clouse 2005).

It has been shown that prolonged hyperglycemia results in overproduction of free radicals, oxidative stress, and inflammation (Matough *et al.* 2012, Tangvarasittichai 2015). Diabetes-induced oxidative stress is characterized by high levels of malondialdehyde (MDA) and also diminished antioxidant enzymes activities (Alipour *et al.* 2012, Naderi *et al.* 2015).

Type 1 diabetes is a lifelong metabolic disorder which is currently treated with a complex regimen of insulin, diet, and exercise (Kakleas *et al.* 2009). Some other therapies for diabetes include various oral antidiabetic agents such as sulfonylureas, biguanides, and glinides (Saxena and Vikram 2004). Controlling diabetes and its complications is still a global problem and fully successful treatments are yet to be discovered (Malviya *et al.* 2010). Recently, phytochemicals identified from traditional medicinal plants are presenting an opportunity for the development of new types of therapeutics (Tiwari and Rao 2002). Natural products with antidiabetic effects have either a direct blood glucose lowering effect or an insulinomimetic property (Patel *et al.* 2012). Among medicinal plants, garlic (*Allium sativum*) has been used in herbal medicine for the treatment of diabetes (Lanzotti 2006). Garlic has several effects such as antihyperglycemic, insulinotropic, cholesterol and lipid-lowering and antithrombotic effects (El-Demerdash *et al.* 2005, Rahman and Lowe 2006, Patel *et al.* 2012). In addition, antioxidant and anti-inflammatory effects of garlic have been reported in different organs (Capasso 2013, Nasr 2014).

The purpose of the present study was to examine the effects of garlic on anxiety- and depression-related behaviors, and oxidative status of the brain in diabetes induced by streptozotocin (STZ).

Materials and methods

Animals

Fifty-six male Wistar rats (230–250 g) were obtained from a colony of Tabriz University of Medical Sciences and

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maintained on a 12 h light–dark cycle in an air-conditioned constant temperature (22 ± 1 °C) room. Food and water were made available *ad libitum*. The Ethics Committee for Animal Experiments at Tabriz University of Medical Sciences approved the study plan, and all experiments were conducted in accordance with the National Institute of Health Guide for the Care and Use of Laboratory Animals.

Animals were randomly divided into seven experimental groups (n = 8) as following: Control (C); diabetic animals that received saline (DS); diabetic animals that received one of the three garlic homogenates doses 0.1, 0.25, and 0.5 g/kg (DG 0.1, DG 0.25, and DG 0.5, respectively); diabetic animals that received diazepam (DD); diabetic animals that received imipramine (DI).

In order to induce diabetes, a single dose (60 mg/kg) of STZ (Sigma Chemical Co., St Louis, MO) was dissolved in saline and administered intraperitoneally (Pushparaj *et al.* 2007). To confirm the induction of diabetes, blood glucose levels were measured using a strip-operated glucometer (Elegance, CT-X10 Germany) 72 h after the STZ injection. Animals with blood glucose levels higher than 300 mg/dl were considered diabetic.

Animals in DS group received 0.9% saline and those in DG groups received garlic homogenates by gavage for 10 days after diabetes induction. Single dose (1 mg/kg) of diazepam (Sigma, Germany), as a positive control for anxiolytic effect, was subcutaneously administered in DD group thirty minutes before behavioral tests (Frassetto *et al.* 2010). Animals in DI group received 15 mg/kg of imipramine (Sigma, Germany), as a positive control for antidepressant effect (Sanmukhani *et al.* 2011). All animals were weighed before and after interventions.

Preparing garlic (Allium sativum) homogenate

Garlic bulbs were purchased from a local market. Cloves were peeled, sliced, ground into a paste, and then suspended in distilled water. The garlic homogenate was prepared freshly each day. Animals were gavaged at different doses (0.1, 0.25, and 0.5 g/kg) of garlic homogenates for ten days after diabetes induction. On the 10th day, behavioral tests for anxiety- and depression-related behaviors were performed between 9 and 11 am in a quiet room by a blinded experimenter to the treatments.

Elevated plus maze

The Elevated plus maze (EPM) as a common behavioral test is validated to assess the anxiolytic effects of pharmacological agents in rodents. The EPM apparatus was a metallic, cross-shaped maze, consisting of four arms arranged in the shape of a plus sign. Two of the arms or doorways had no side or end walls (open arms; 50 × 10 cm). An edge of 1 cm in height had been installed on both sides and an open doorway. The other two arms had side walls and end walls, but were open on the top (closed arms; 50 × 10 × 40 cm). On the intersection of arms, there was a square platform of 10 × 10 cm. The maze was elevated to a height of 50 cm.

Animals were individually placed in the center of the maze facing an open arm and allowed 5 min of free exploration. All sessions were videotaped and the number of entries into the open and closed arms and the total time spent in the open and closed arms were measured. Entry was defined as all four paws in the arms. The percentage of open arm entries and open arm time as the standard anxiety indices were calculated as follows: (a) %OAT (the ratio of times spent in the open arms to total times spent in any arms*100); (b) %OAE (the ratio of entries into open arms to total entries*100); (c) TAE (total arm entries). Total arm entries were considered as the index of locomotor activity (Farajdokht *et al.* 2017b).

Forced swimming test

A two-day procedure forced swimming test (FST) was used for the evaluation of the antidepressant effect of *Allium sativum* in rats. Animals were placed in Plexiglas cylindrical containers (35 × 45 cm) during FST. The containers were filled with water at 23 ± 1 °C with a 30-cm depth. FST was performed 60 min after treatments.

On the first day, the rats were forced to swim for 15 min under a condition in which escape was not possible. On the second day, the duration of the test was 6 min and immobility time was measured during the last 4 min. Immobility was defined as no additional activity except the movements which were required to keep the rat's head above the water (Hosseinzadeh *et al.* 2007).

Determination of the brain MDA levels and antioxidant enzyme activity (GPX and SOD)

At the end of behavioral tests, animals were deeply anesthetized with a mixture of ketamine (80 mg/kg, i.p) and xylazine (10 mg/kg, i.p), then sacrificed and brain tissues were removed and kept in −70 °C for further analysis. In order to assess the antioxidant enzyme activities (SOD and GPx), brain tissue samples were homogenized in 1.15% KCl solution and centrifuged at 5000 rpm for 10 min at 4 °C for obtaining the supernatant. The MDA levels, as a marker of oxidative stress, were determined using the thiobarbituric acid reactive substances (TBARS) method and expressed as nmol/mL (Farajpour *et al.* 2017c). SOD and GPx activities in the supernatants were determined using RANSOD kits (Randox Laboratories Ltd, United Kingdom) according to the manufacturer's instructions and expressed as IU/mg protein in tissue homogenate (Farajdokht *et al.* 2017a, Pourmemar *et al.* 2017).

Statistical analysis

Data were statistically analyzed by SPSS software (version 16). Data were analyzed by one-way analysis of variance (ANOVA) followed by Tukey's *post-hoc* test. The significance level was set at $p < .05$. Results are expressed as Mean ± SEM.

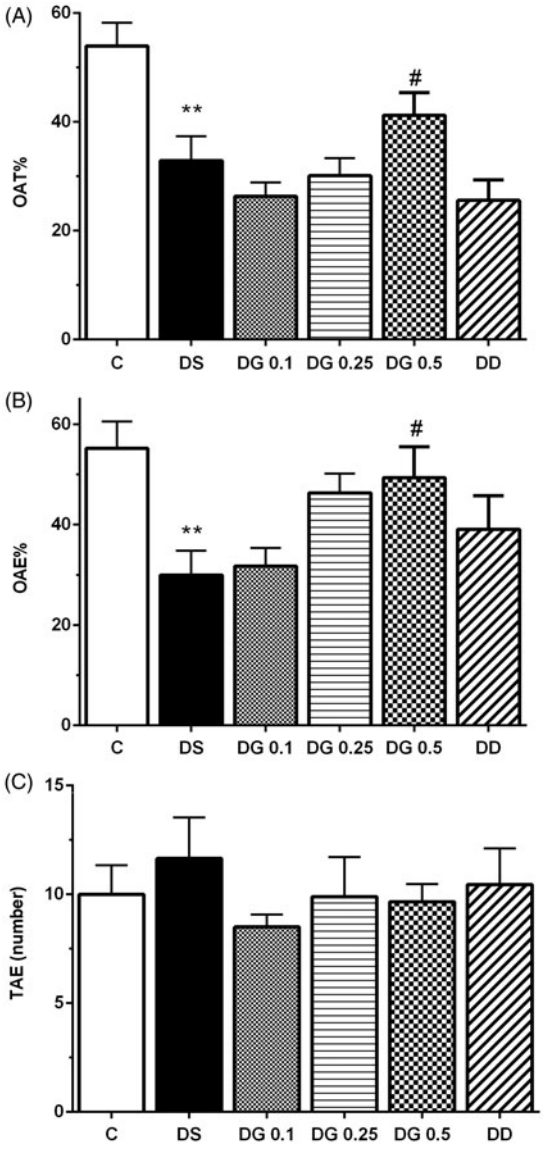


Figure 1. The effects of different doses of garlic on the anxiety-related behaviors in the EPM: (A) open arm time percent (OAT %), (B) open arm entrance percent (OAE %) and (C) total entries (TAE). Data are expressed as Means ± SEM ($n = 8$). One-way ANOVA followed by Tukey's *post hoc* test: $^{**}p < .01$ vs. control group, $\#p < .05$ vs. DS group. [C: control; DS: diabetic + saline; DG 0.1: diabetic + garlic (0.1, 0.25, and 0.5 g/kg); DD: diabetic + diazepam].

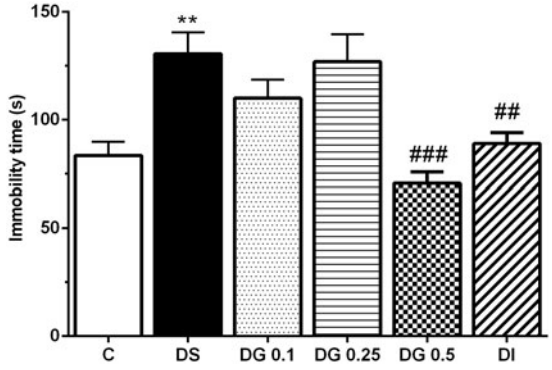


Figure 2. The effects of different doses of garlic on depression-related behaviors in diabetic rats in FST. Data are expressed as Mean ± SEM ($n = 6$–8). One-way ANOVA followed by Tukey's *post hoc* test: $^{**}p < .01$ vs. control group, $\#\#p < .01$, $\#\#\#p < .001$ vs. DS group. [C: control; DS: diabetic + saline; DG 0.1: diabetic + garlic (0.1, 0.25, and 0.5 g/kg); DI: diabetic + imipramine].

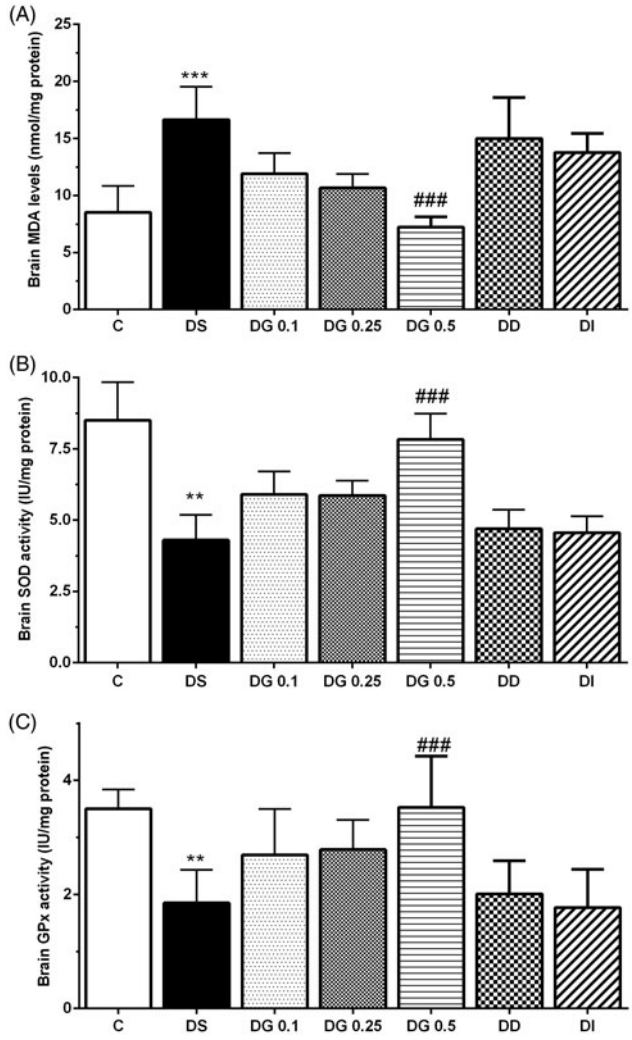


Figure 3. The effects of different doses of garlic on (A) Mean malondialdehyde (MDA) values, (B) superoxide dismutase (SOD) and (C) glutathione peroxidase (GPx) in different groups. Data are expressed as Mean ± SEM ($n = 6$–8). One-way ANOVA followed by Tukey's *post hoc* test: $^{**}p < .01$ and $^{***}p < .001$ vs. control group, $\#\#\#p < .001$ vs. DS group. [C: control; DS: diabetic + saline; DG 0.1: diabetic + garlic (0.1, 0.25, and 0.5 g/kg); DD: diabetic + diazepam; DI: diabetic + imipramine].

Results

The effect of garlic on anxiety-related behaviors

The results of EPM test revealed that induction of diabetes significantly ($p < .01$) decreased OAT% and OAE% ($p < .01$) in the DS group as compared to the control group. Nevertheless, garlic 0.5 g/kg treatment in diabetic rats caused a significant ($p < .05$) increase in OAT% and OAE% as compared to DS group (Figure 1(A,B)). However, single i.p injection of diazepam did not significantly change OAT% or OAE% parameters in the diabetic animals. Moreover, the results of total entrance to open and closed arms, index of locomotor activity, showed no significant differences among different groups (Figure 1(C)).

The effect of garlic on depression-related behaviors

As shown in Figure 2, immobility time was significantly ($p <$ .01) increased in the DS group when compared to the

Table 1. The effect of ten days of garlic treatment (0.1, 0.25, and 0.5 g/kg) on body weight (BW) in STZ-induced diabetic rats.

	BW before	BW after	*p* Value
C	231 ± 4.5	271 ± 1.5	<.05
DS	241.25 ± 2.95	213.125 ± 3.64	<.001***
DG 0.1	228.71 ± 2.63	212.14 ± 2.86	>.05
DG 0.25	227.25 ± 2.51	224.62 ± 3.01	.514
DG 0.5	235.5 ± 3.8	249 ± 3.15	<.05#
DD	241 ± 2.98	216.125 ± 2.40	>.05
DI	242.5 ± 2.56	218.28 ± 3.56	>.05

Data are presented as means ± SEM Paired sample *t*-test. BW: body weight; C: control; DS: diabetic + saline; DG 0.1: diabetic + garlic (0.1, 0.25, and 0.5 g/kg); DI: diabetic + imipramine.
***$p < .001$; #$p < .05$.

control group. However, treatment with garlic 0.5 mg/kg markedly ($p < .001$) decreased the total duration of immobility compared to DS group (Figure 2). Imipramine (15 mg/kg) treatment also significantly ($p < .01$) decreased immobility time in the FST as compared to the DS group.

The effect of garlic on oxidative stress status in the brain

Our results also demonstrated a significant ($p < .001$, Figure 3(A)) increase in MDA levels and a significant ($p < .01$, Figure 3(B,C)) decrease in the SOD and GPx enzyme activities in the brain tissue of DS group when compared with control group. However, garlic (0.5 g/kg) treatment significantly ($p < .001$) attenuated MDA levels and increased SOD and GPx activities as compared to the DS group.

The effect of garlic on body weight in diabetic rats

The effect of 10 days of garlic treatment on body weight was also investigated in the diabetic rats. Our result showed that induction of diabetes significantly ($p < .001$) reduced the body weight of the DS group. However, garlic treatment (0.5 g/kg) significantly ($p < .05$) prevented weight loss after induction of diabetes with STZ. Treatment with imipramine or diazepam did not significantly change the body weight when compared with DS group (Table 1).

Discussion

Our finding revealed that diabetes induction produced a characteristic anxiety-related behavior, confirmed by decreased OAT% and OAE% in the EPM. The results of FST also demonstrated that diabetes significantly increases immobility time. Whereas, garlic (0.5 g/kg) treatments significantly increased OAT% and OAE% in the EPM, and decreased immobility time in the FST, suggesting anxiolytic and antidepressant-like activities

Previous studies showed that diabetes and metabolic disorders are associated with prominent anxiety symptoms and increased the possibility of having anxiety disorders (Fisher *et al.* 2008, Smith *et al.* 2013, Bayandor *et al.* 2018). In the present study, results of EPM test showed that diabetic animals displayed more time spent in the enclosed arms and had more entries into the closed arms which were significantly attenuated by garlic (0.5 g/kg) treatment. According to the previous report, the increase in two parameters, OAT%, and OAE%, indicates the anxiolytic effect (Kesmati *et al.* 2014). Therefore, the present study for the first time indicates that garlic possesses active anxiolytic effects. In contrast to Frassetto *et al.* (2010) study, in the current study, no significant change was observed in anxiety-related behaviors of the diazepam-treated group. In line with our results, López-Rubalcava *et al.* (2013) reported that the response to diazepam was reduced in diabetes induced by streptozotocin in mice. Another study also found that that diazepam (1 mg/kg) does not have an anxiolytic effect in the diabetic rats (Ramanathan *et al.* 1998)

In the FST immobility time is an index of despair and depression-related behaviors, and agents that decrease the immobility time have antidepressant effects (Yankelevitch-Yahav *et al.* 2015). In this study, diabetic animals (saline group) also demonstrated longer immobility time in the FST than the control group; however, raw homogenized garlic (0.5 g/kg) significantly decreased immobility time as compared to saline group indicating that garlic has an antidepressant effect in diabetes. In line with our findings, Dhingra and Kumar (2008) showed that garlic extract has an antidepressant-like activity in mice by inhibiting brain monoamine oxidases (MAO-A and MAO-B) activities. The antidepressant effect of dietary supplement garlic has also been shown in ovariectomized female rats (Bhandare and Tembhurne 2014).

Oxidative stress is a common complication of diabetes. Free radicals are formed disproportionately in diabetes by glucose oxidation, nonenzymatic glycation of proteins, and the subsequent oxidative degradation of glycated proteins (Khan *et al.* 2015). Recent studies have shown direct involvement of oxidative stress in anxiety-and- depression-related behaviors of rats (Salim *et al.* 2010). Moreover, disturbances in the metabolism of glucose and oxidative stress are considered to be the main factors in the pathophysiology of diabetes-associated psychiatric disorders such as depression and anxiety (de Morais *et al.* 2014, Khan *et al.* 2015). Studies have also shown that whole-body glucose disposal is reduced in patients with typical and atypical depression (Schweiger *et al.* 2008). A good strategy for glycemic control may contribute to improvements in mood (Lustman and Clouse 2005). Moreover, several studies indicated that depression and anxiety disorders are related to lowered antioxidative enzyme defense, and increased reactive oxygen species (ROS) and MDA levels (Krolow *et al.* 2014, Balmus *et al.* 2016). Additionally, it has been suggested that phytochemicals with antioxidant properties are able to alleviate diabetic complications (Dembinska-Kiec *et al.* 2008). In the present study, in the diabetic animals, MDA levels significantly increased possibly as a result of hyperglycemia and oxidative stress induction. Nevertheless, oral garlic treatment reduced MDA levels and enhanced enzymatic antioxidant activity (SOD, GPx) in the brain which was accompanied by attenuation of depression and anxiety-related behaviors in the diabetic rats. The antidiabetic effect of raw garlic has been well proven in the type 1 diabetes model (Banerjee *et al.* 2002,

Eidi *et al.* 2006). Moreover, different preparations of garlic, mainly aged garlic extract (AGE), have been shown to have promising antioxidant potential (Elosta *et al.* 2017). Given garlic contains antioxidant compounds such as S-allyl cysteine; it has a capacity to scavenge free radicals (Chung 2006). Indeed, garlic consumption improved the balance between brain oxidants and antioxidants markers. In line with our finding, several studies have reported that garlic extract increases total serum antioxidant levels in diabetic rats (Al-Qattan *et al.* 2008, Lee *et al.* 2009, Drobiova *et al.* 2011). Therefore, it seems that garlic attenuated diabetes-induced depression and anxiety through its antioxidant effects.

Our findings also showed that diabetes significantly decreased body weight (BW) in the experimental groups, and garlic (0.5 g/kg) treatment for ten days reduced body weight loss in the diabetic animals. This finding is in agreement with the findings of Bokaeian *et al*. (2010) which concluded that administration of garlic extract reduced hyperglycemia, polydipsia, polyphagia, and associated weight loss of STZ- treated rats. It has also reported that long-term treatment of diabetic animals with garlic oil can improve oral glucose tolerance (Liu *et al.* 2006). This effect of garlic on body weight may result from the effect of garlic on the entry of glucose into the cells and increase the sensitivity of cells to insulin and, in turn, increases in glucose metabolism (Dhingra and Kumar 2008).

Conclusion

Overall, our findings indicate that raw garlic homogenate improves anxiety- and depression-related behaviors in diabetic rats possibly through attenuation of oxidative stress and augmentation the antioxidant defense against diabetes.

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Disclosure statement

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